**ALL RTU EXAM QUESTIONS LAST 4 YEARS (UNIT 1, 2)**

**INFORMATION THEORY CODING (V SEM CSE)**

**QUESTION BANK (UNIT 1, 2)**

**PART A**

Q1: Prove the the following statement "The sun rises in the east" contains zeo amount of information. **(RTU 2024)**

Q2: Show that if there is "n" number of equally likely message then entropy of the source is $\log_2 n$. **(RTU 2024)**

Q3: Define Kraft inequality. **(RTU 2024)**

Q4: Explain prefix code with example. **(RTU 2024, 2021 )**

Q5: Explain the sufficient & necessary condition for code to be instantaneous. Give the Example **(RTU 2023)**

Q6: Explain the properties of mutual information. **(RTU 2023)**

Q7: What is perfect code? **(RTU 2023)**

Q8: Define term uncertainty and information. **(RTU 2023)**

Q9: Prove that the following statement "If receiver knows the message being transmitted the amount or information carried is zero" **(RTU 2022)**

Q10: State the Shanon's first theorem for source coding. **(RTU 2022)**

Q11: Explain channel coding theorem for discrete memoryless channel. **(RTU 2021)**

Q12: Explain channel capacity theorem. **(RTU 2021)**

Q13: Define mutual information. **(RTU 2021)**

**PART B**

Q1: A Source Produces one of five possible message during each interval having Probabilities $P_1$=1\2, $P_2$ = ¼, $P_3$= 1/8, $P_4$= 1/8, $P_4$= 1/16.Determine the entropy and information rate if there are 16 outcome per second. **(RTU 2024)**

Q2: Show that for discrete binding channel: **(RTU 2024,2021)**

H( X, Y ) = H(X | Y)+ H(Y)

H( X, Y ) = H(Y | X)+ H(X)

Q3: For a discrete memoryless source 'x' with four symbols $x_1, x_2, \ldots\ldots\ldots x_4$. Find a compact code usig Shanon-Fano method and also find efficiency: **(RTU 2024)**

$P(x_1)=1/2$ $P(x_2)=1/4$ $P(x_3)=1/8$ $P(x_4)=1/8$

Q4: A voice grade channel of telephone network has a bandwidth of 3.5khz. Calculate the information capacity of telephone channel for signal to noise ratio 30 db. **(RTU 2024)**

Q5:Consider a source with 8 alphabets with probabilities given below: **(RTU 2023)**

| [x]= x1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|
| [p]=1/4 | 0.2 | 0.18 | 0.15 | 0.12 | 0.08 | 0.05 | 0.02 |

Contruct code using Shanan Fano & calculate the efficiency.

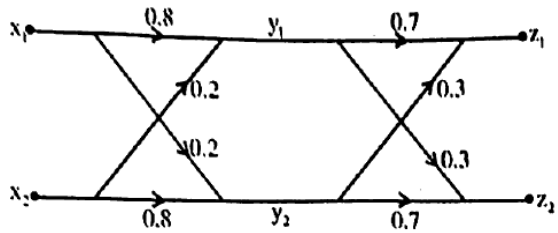
Q6: A communication system employs a continuous source. The channel noise is white and Gaussian. The bandwidth of source output is 10 MHZ and signal to noise ratio is 100. What will be the channel capacity? **(RTU 2023)**

Q7: State and establish Kraft inequality. : **(RTU 2023)**

Q8: Derive the expression for channel capacity when bandwidth become infinite. **(RTU 2023)**

Q9: Prove that the upper boundary on entropy is given as $H_{max} \leq \log_2 M$ here M is the number of message emmited by the source. " **(RTU 2022)**

Q10:The BSC are connected in cascade as shown in fig. **(RTU 2022)**




(a) Determine the transition matrix for discrete memoryless channel.

(b) Determine $P(z_1)$ & $P(z_2)$ if $P(x_1) = 0.6$ , $P(x_2)=0.4$

Q.11 Consider a source S=[S1, S2] with probabilities ¾ and ¼ respectively. Obtain Shannon-Fano code for sourcres . Calculate efficiency for each case. **(RTU 2021)**

Q.12 What is coding efficiency? Show that coding efficiency is maximum when P(0)=P(1). **(RTU 2021)**

Q13: Write a short notes on noise free channel & channel's theorem. **(RTU 2021)**

**PART C**

Q1: a) Discuss the concept of joint entropy and mutual information. **(RTU 2024)**

b) Explain different types of channels with their channel matrix. **(RTU 2024)**

Q2: Construct Huffman code for following set of messages. Also find the efficiency. **(RTU 2024)**

$P(x_1)=0.49$ $P(x_2)=0.14$ $P(x_3)= 0.14$ $P(x_4)= 0.07$ $P(x_5)= 0.07$ $P(x_6)= 0.04$

$P(x_7)= 0.02$ $P(x_8)= 0.02$ $P(x_8)= 0.01$

Q.3 Joint probability matrix of a discrete channel is given by: **(RTU 2023)**

$$P(X,Y) = \begin{matrix} 0.05 & 0.05 & 0.02 & 0.05 \\ 0.15 & 0.16 & 0.01 & 0.09 \\ 0.12 & 0.03 & 0.02 & 0.05 \\ 0.01 & 0.12 & 0.01 & 0.06 \end{matrix}$$

Compute marginal, conditional and joint entropies and verify their relation.

Q4: For a discrete memoryless source 'x' with six symbols $x_1, x_2, \ldots\ldots\ldots x_6$. Find a compact code for every symbol if the probability distribution is as follows: **(RTU 2022)**

$P(x_1)=0.3$ $P(x_2)=0.25$ $P(x_3)=0.2$

$P(x_4)=0.12$ $P(x_5)=0.08$ $P(x_6)=0.05$